

Code: BCS606**Course: Distributed Systems****Credits: 3****L:T:P 3:0:0****SEE: 50 Marks****CIE: 50 Marks****SEE Hours: 3****Max. Marks: 100**

Prerequisites if any	Nil
Learning objectives	1. Learn the fundamentals of distributed systems, through examples 2. Learn to use appropriate remote invocation techniques for communication in DS

Course Outcomes: *On the successful completion of the course, the student will be able to*

COs	Course Outcomes	Bloom's level
CO1	Explain the applications and challenges arising from the construction of distributed systems	Understanding
CO2	Make use of appropriate remote invocation technique to communicate in distributed systems	Apply
CO3	Use distributed objects and components to develop applications using CORBA middleware	Apply
CO4	Examine algorithms for clock synchronization and use appropriate concurrency control algorithm for transactions	Analyze

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	-	-	-	-	-	-	-	-	3	1	-
CO2	3	3	-	-	-	-	-	-	-	-	-	3	-	2
CO3	3	2	2	-	2	-	-	-	-	-	-	3	2	-
CO4	3	2	3	2	2	-	-	-	-	-	-	3	2	2

Mapping with POs and PSOs:**Mapping Strength: Strong– 3 Medium – 2 Low – 1**

Course Structure

Sl. No	Modules	No. of Lecture Hours	No. of Tutorial Hours	No. of Practical Hours
Module – 1: Characterization of Distributed Systems, System models				
1.1	Introduction to DS, examples	1	0	0
1.2	Trends in distributed system. Focus on resource sharing	2	0	0
1.3	Challenges	1	0	0
1.4	Physical models, Architectural Models, Fundamental Models	3	0	0
1.5	Case study: WWW	1	0	0
Module – 2: Inter Process Communication, Remote Invocation				
2.1	Introduction, API for Internet Protocols	1	0	0
2.2	External data representation and Marshalling	2	0	0
2.3	Client – Server Communication	1	0	0
2.4	RR Protocol, RPC, RMI	3	0	0
2.5	Case study: SUN RPC	1	0	0
Module – 3: OS Support, Distributed Objects, DFS				
3.1	Processes, Threads, OS Architecture	2	0	0
3.2	Introduction to Distributed Objects and components	1	0	0
3.3	Case study: CORBA – architecture, service	2	0	0
3.4	Distributed file systems, File service architecture	2	0	0
3.5	Case study: Sun Network File System	1	0	0
Module – 4: Time and Global State				
4.1	Clocks, Events and process status	2	0	0
4.2	Synchronizing physical clocks	2	0	0
4.3	Logical time and logical clocks, Global states	3	0	0
4.4	Distributed debugging	1	0	0
Module – 5: Transactions and Concurrency Control				
5.1	Introduction to Transactions, Nested Transactions	3	0	0
5.2	Locks, Optimistic Concurrency Control, Timestamp ordering	4	0	0
5.3	Comparison of methods for concurrency control	1	0	0
Total No. of Lecture Hours		40	-	-
Total No. of Tutorial Hours			00	-
Total No. of Practical Hours				00

Textbook:

1. George Coulouris, Jean Dollimore, Tim Kindberg, Gordon Blair: Distributed Systems – Concepts and Design, Fifth Edition, Pearson Publications, 2012.

Reference Book:

1. Maarten van Steen. Andrew S. Tanenbaum: Distributed Systems, Third edition, 2017

Online Resources:

1. www.cdk5.net/corba
2. <https://www.coursera.org/specializations/pcdp>
3. https://onlinecourses.nptel.ac.in/noc21_cs87/